Internet Of Things.

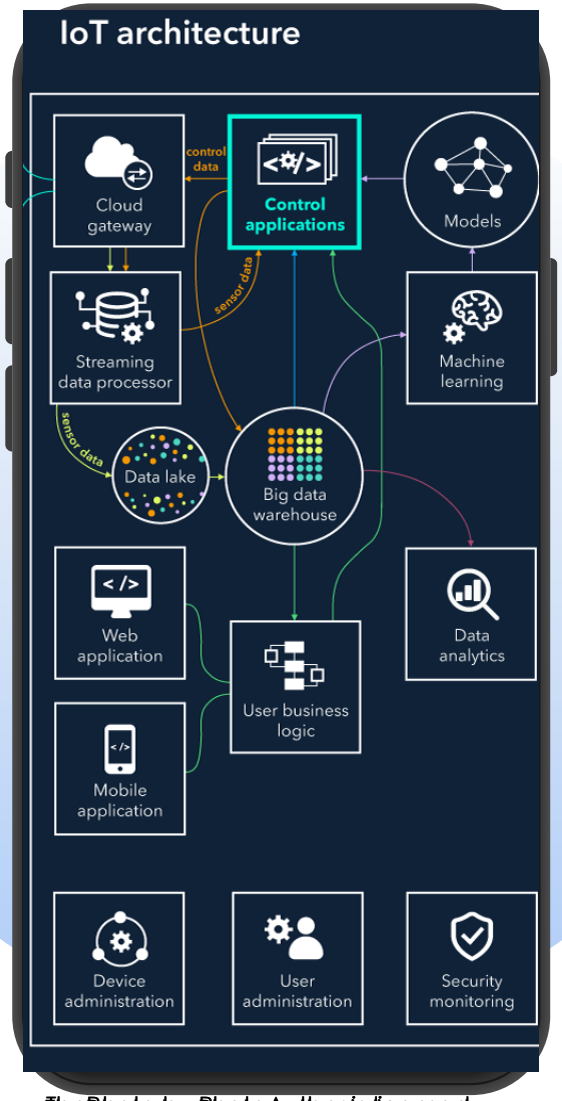
The future.

Agenda

- What is iot?
- Key Components of iot
- How it works
- Advantages
- Disadvantages
- Application of iot
- Conclusion

What is IOT?

IoT (Internet of Things) means everyday objects like lights, watches, cars, or machines are connected to the internet so they can share data and work smartly without needing humans to control them all the time. In short, it’s about making devices “talk” to each other to make life easier.



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Key Components

- **Sensors** provide the raw input.
- **Connectivity** ensures communication.
- **Data processing** extracts value.
- **User interface** delivers insights to people.

How it Works?

The Internet of Things (IoT) works by connecting everyday physical devices—like appliances, vehicles, and machines—to the internet using sensors, software, and communication technologies. These devices collect and share data in real time, enabling automation, monitoring, and smarter decision-making without direct human input.

Advantages of IOT

1. **Automation & Efficiency**
2. **Real-Time Monitoring**
3. **Cost Reduction**
4. **Enhanced Decision-Making**
5. **Improved Safety & Security**

Disadvantages of IOT

1. Privacy & Security Concerns
2. System Complexity
3. Dependability & Reliability
4. Job Displacement
5. Job Displacement

Applications of IOT

1. Smart Homes

- **Security systems**: IoT-enabled locks, cameras, and alarms.

2. Healthcare

- **Remote patient monitoring**: Wearables track heart rate, glucose levels, and sleep.

3. Agriculture

- **Precision farming**: Sensors monitor soil moisture, crop health, and weather

4. Manufacturing & Industry (IIoT)

- **Supply chain optimization**: Real-time tracking of goods and materials.

5. Smart Cities

- **Traffic management**: IoT sensors reduce congestion and improve safety.
- **Energy grids**

Challenges faced in IOT

The challenges of IoT can be understood as a set of intertwined problems that arise when billions of devices are connected and constantly exchanging information. One of the most pressing issues is security. Many IoT devices are designed with limited computing power and minimal protection, which makes them vulnerable to hacking. Once compromised, these devices can be used to access sensitive networks or personal data. Closely tied to this is the problem of privacy, since IoT systems collect vast amounts of information about people’s habits, health, and environments. Ensuring that this data is not misused or exposed is a constant struggle.

Summary

"The Internet of Things is transforming the way we live and work, connecting devices, people, and processes in ways that were unimaginable a decade ago. While it brings enormous opportunities for efficiency, innovation, and convenience, it also raises serious challenges around security, privacy, scalability, and regulation. The future of IoT depends on how well we address these challenges—through stronger standards, smarter technologies, and responsible practices. If we succeed, IoT will not just connect devices, but truly connect societies, driving progress toward a more intelligent and sustainable world."

Thank you